

CPSC 416 Distributed Systems

Winter 2023 Term 1 (September 12, 2023)

Tony Mason (fsgeek@cs.ubc.ca), Lecturer



Logistics



Teaching Assistants

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Office Hours

Remember: Use Piazza for **all** official course-related communications

- Not on Piazza? Not official.
- Canvas “comments/messages” **are not monitored**



Office Hours:

Who	When	Where
Tony	Monday 14:00-15:00 Wednesday 16:00-17:00	Discord
Andy	By appointment	TBA
Hamid	Friday 16:30-18:00	Kaiser 4075
Jonas	Thursday 11:00-12:30	X150, Table 1&2
Cathy	Friday 09:00-10:30 (Starting Sep. 22)	X237

Self-Assessment

This week

- Post-lecture review assignment (Useful?) – Due before *next* lecture
- Distributed Systems Design Recipe assignment – Due Friday (Sep 15 @ 23:59)



Note:

- You are strongly encouraged to collaborate with others on this
- You should use tools at your disposal to answer these questions
 - **Including your favourite large language model** (e.g., ChatGPT, your older sibling, or your pet.)
- As previously noted, you get full credit if you submit. **Do not forget to submit it.**

Failure



Today's Failure

Source: [Microsoft had three staff at Australian data centre campus when Azure went out - Cloud - Storage – iTnews](#)

When: 08:41 UTC on 30 August 2023 to 06:40 UTC on 1 September 2023

Duration: 46:59 (e.g, just short of **two days**)

Effect: “Issues affecting access to” Azure, Microsoft 365 and Power Platform.

TL;DR version: power sag caused chiller failure caused thermal failure of storage devices. Secondary problems (“bugs”) caused approval of stale requests and marked operational nodes as “not healthy” (degraded or failed). Recovery tools didn’t work well (“SQL did not have well tested tools on hand that were built to move databases when the source ring was in [a] degraded health scenario”)

Cascading failures are common. Recovery is hard – seldom used paths, staff training, etc.



MURPHY
was an optimist



Learning Goals



Learning Goals

Chapter 3: File Formats

- How you organize data matters (performance, recovery)
- Details matter (Understanding trade-offs)
 - Media characteristics
 - Usage patterns



Chapter 4: Implementing B-Trees (Organizing persistent storage)

- How do we handle mutability efficiently?
- How do we access data quickly?
- How do we deal with space reclamation?

Note: these issues are recurring themes in systems.

Discussion



Discussion

Too much logistics and failure... Let's talk.

Petrov, Chapters 3 & 4



Chapter 3: High Level

Cell Layout and Data Types

Slotted Pages

Variable Size Data

B-Tree Layout and Pages

Versioning and File formats

Checksumming and Data Integrity



Chapter 4: High Level

B-Trees: key organizational structure for storage

Search

Splitting and Merging

Maintaining Balance

Optimized Data Loading

Space Efficiency

- Compression
- Cleaning/Compaction



Questions?





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